EX NO:01	**Tableau Aggregation** (Sum, Min, Max, Average, Count, Count Distinct, Median, Variance, Population Variance, Standard Deviation, and Population Standard Deviation)

AIM:

To perform various data aggregation operations in Tableau using Sum, Min, Max, Average, Count, Count Distinct, Median, Variance, Population Variance, Standard Deviation, and Population Standard Deviation and visualize the results for different regions.

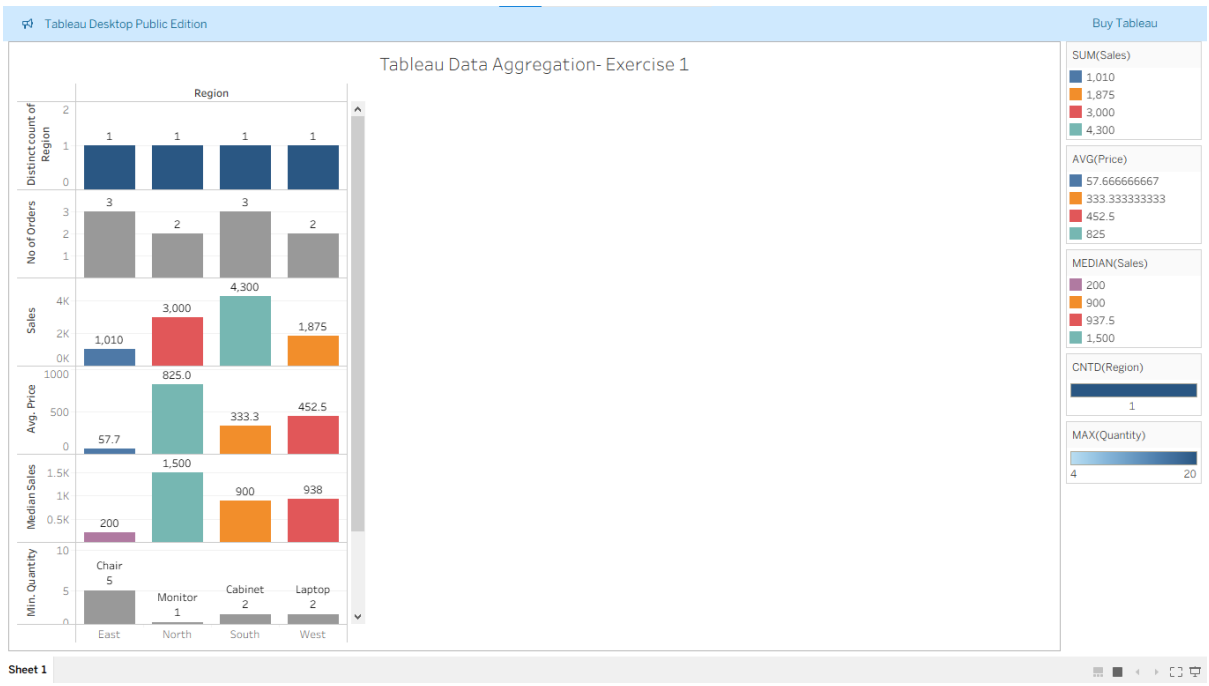
PROCEDURE:

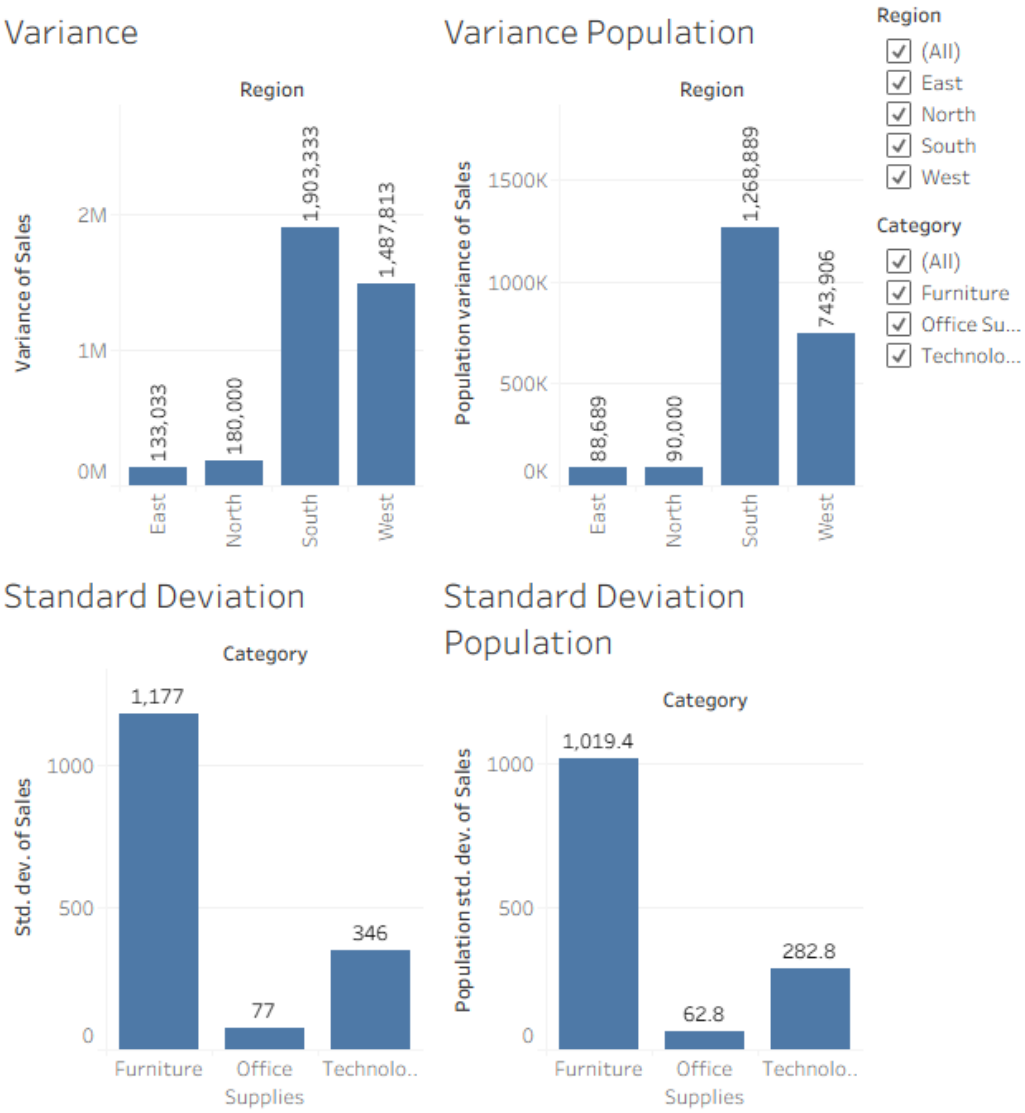
Step 1: Start the process.
Step 2: Open **Tableau Public**. In the **Connect** pane, click on **Text File**, select the sales_dataex1 dataset, and load it into Tableau.
Step 3: Go to **Sheet 1**. Drag **Region → Columns** and **Sales → Rows**. Tableau automatically applies **SUM(Sales)**. Set the sheet title as *Sales – Sum*.
Step 4: Go to **Sheet 2**. Drag **Region → Columns** and **Quantity → Rows**. Right-click on **Quantity → Measure → Minimum**. Set the sheet title as *Quantity – Minimum*.
Step 5: Go to **Sheet 3**. Drag **Region → Columns** and **Quantity → Rows**. Right-click on **Quantity → Measure → Maximum**. Set the sheet title as *Quantity – Maximum*.
Step 6: Go to **Sheet 4**. Drag **Region → Columns** and **Price → Rows**. Right-click on **Price → Measure → Average**. Set the sheet title as *Average (Price)*.
Step 7: Go to **Sheet 5**. Drag **Region → Columns** and **Order ID → Rows**. Right-click on **Order ID → Measure → Count**. Set the sheet title as *Order ID – Count*.
Step 8: Go to **Sheet 6**. Drag **Region → Columns** and **Region → Rows** again. Right-click → **Measure → Count Distinct**. Add **Region** to **Label** and **Color**. Set the sheet title as *Region – Count Distinct*.
Step 9: Go to **Sheet 7**. Drag **Region → Columns** and **Sales → Rows**. Right-click on **Sales → Measure → Median**. Set the sheet title as *Sales – Median*.
Step 10: Go to **Sheet 8**. Drag **Region → Columns** and **Sales → Rows**. Right-click on **Sales → Measure → Variance**. Set the sheet title as *Variance of Sales*.
Step 11: Go to **Sheet 9**. Drag **Region → Columns** and **Sales → Rows**. Right-click on **Sales → Measure → Variance (Population)**. Set the sheet title as *Population Variance of Sales*.
Step 12: Go to **Sheet 10**. Drag **Category → Columns** and **Sales → Rows**. Right-click on **Sales → Measure → Standard Deviation**. Set the sheet title as *Standard Deviation of Sales*.
Step 13: Go to **Sheet 11**. Drag **Category → Columns** and **Sales → Rows**. Right-click on **Sales → Measure → Standard Deviation (Population)**. Set the sheet title as *Population Standard Deviation of Sales*.
Step 14: Go to **Dashboard 1**. Arrange all **11 sheets** (Sum, Min, Max, Average, Count, Count Distinct, Median, Variance, Population Variance, Standard Deviation, and Population Standard Deviation) neatly in the dashboard layout.
Step 15: Save the workbook at the following location: C:\Users\ADMIN\Documents\My

Tableau Repository\Workbooks

Step 16: View the final output and stop the process.

OUTPUT:





RESULT:

Thus, the above program is executed successfully and verified.

EX NO:02	**Quick Table Calculation** (Percentage of Total, Running Total, Average, Difference, Rank)

AIM:

To visualize various data and table calculations in Tableau.

PROCEDURE:

Step 1: Start the process.

Step 2: Open Tableau Public. In the Connect pane, click on Text File, then select the

sales_dataex1 dataset. Inspect the dataset to check its values and data types.

Step 3: Go to Sheet 1. Drag Sales to the Rows pane and Date to the Columns pane.

Step 4: Right-click on the Date field in the Columns pane and change it to Day.

Step 5: Right-click on Sales in the Rows pane, select Quick Table Calculation, choose Running Total, and set the label.

Step 6: Go to Sheet 2. Drag Category to the Columns pane and Sales to the Rows pane. Right-click Sales, select Quick Table Calculation, choose Percentage of Total, and set the label.

Step 7: Go to Sheet 3. Drag Date to the Columns pane and Sales to the Rows pane. Right-click Date and change it to Month. Then right-click Sales, select Quick Table Calculation, and choose Moving Average. Again, right-click Sales, select Edit Table Calculation, click Average, change the previous value to 3, and set the label.

Step 8: Go to Sheet 4. Drag Date to the Columns pane and Sales to the Rows pane. Right-click Date and change it to Month. Then right-click Sales, select Quick Table Calculation, and click Difference.

Step 9: Go to Sheet 5. Drag Category to the Columns pane and Sales to the Rows pane. Right-click Sales, select Quick Table Calculation, choose Rank, go to Edit Table Calculation, change to Ascending Order, and set the label.

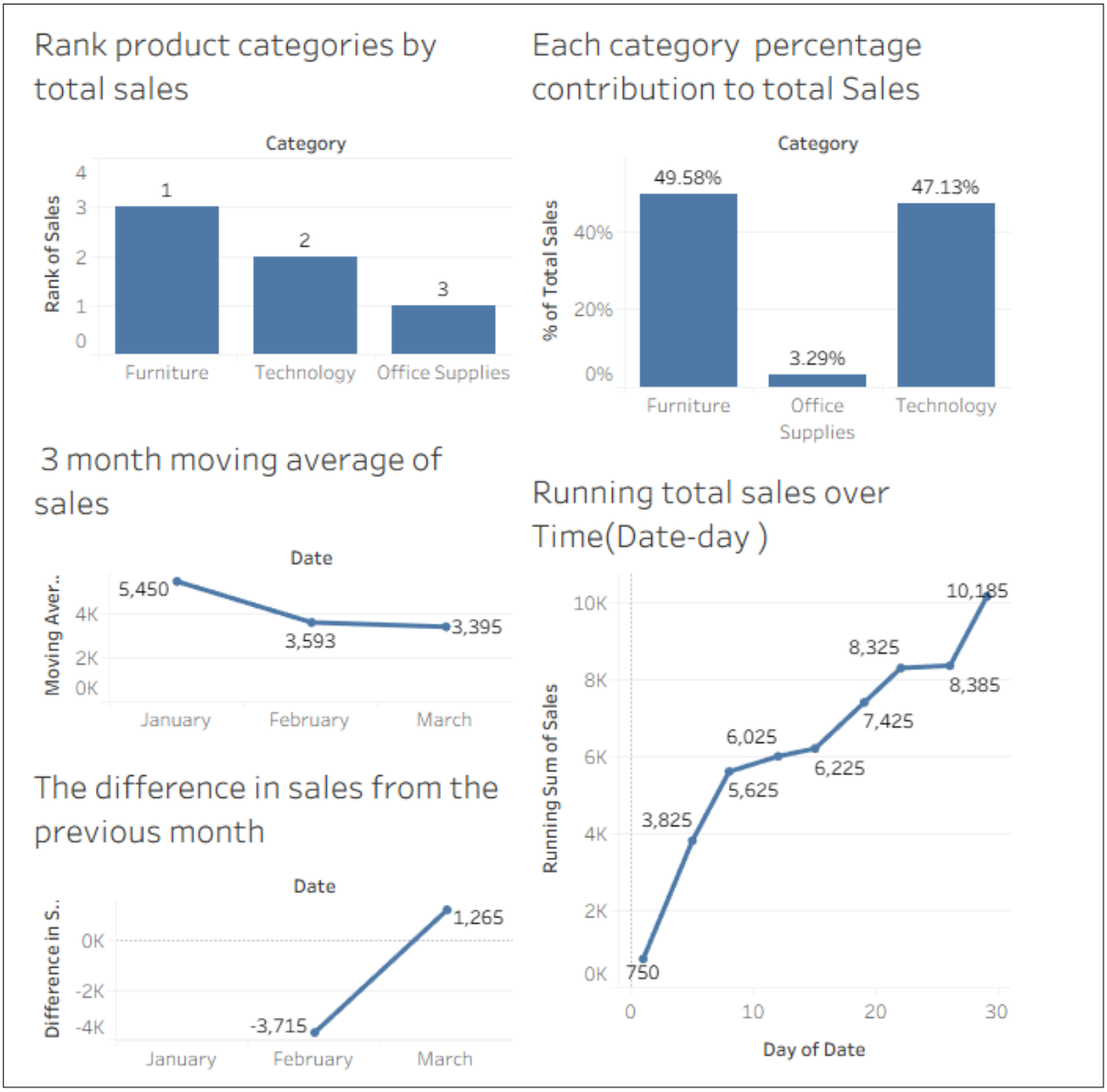
Step 10: Go to Dashboard 1 and place all five sheets accordingly.

Step 11: Save the project in the following location:

C:\Users\ADMIN\Documents\My Tableau Repository\Workbooks

Step 12: View the output and stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:03	**Tableau Calculations** (Calculated Field, Operators, IF–ELSE, Rank, String, Functions)

AIM:

To analyze employee sales performance data in Tableau Public using calculated fields and visualizations to evaluate sales achievement, performance, and compensation.

PROCEDURE:

Step 1: Start the process.

Step 2: Open Tableau Public. In the Connect pane, click on Text File, then select employee_sales_dataex4. Inspect the dataset to check its values and data types.

Step 3: Go to Sheet 1.

Step 4: Drag Employee Name to Columns.

Step 5: Sort them in Descending order.

Step 6: Drag Sales Amount to Rows.

Step 7: Create a calculated field for Year of Service.

DATEDIFF('year', [Join Date], TODAY())

Step 8: Create a calculated field and name it Met Target?

IF [Sales Amount] >= [Target Amount] THEN 'Yes'
ELSE 'No' END

Step 9: Drag Region, Department, and Met Target? to Filters.

Step 10: Click the dropdown menu and select Show Filter.

Step 11: Drag Met Target? to Color and Sales Amount to Text.

Step 12: Drag Employee Name to Columns.

Step 13: Create a calculated field and name it Total Compensation.

[Sales Amount] - ([Sales Amount] - [Bonus])

Step 14: Drag Total Compensation to Rows.

Step 15: Drag High Performer North and Measure Names to Columns.

Step 16: Create a calculated field for Performance.

[Sales Amount] - [Target Amount]

Step 17: Drag Measure Values to Rows and Names to Color.

Step 18: Drag Employee Name to Columns.

Step 19: Create a calculated field for Rank.

RANK(SUM([Sales Amount]))

Step 20: Drag Rank to Rows.

Step 21: Click the dropdown menu from Rank, click Edit Table Calculation, then select Table (Across).

Step 22: Drag Employee Name to Columns.

Step 23: Create a calculated field for Bonus.

[Sales Amount] * 0.05

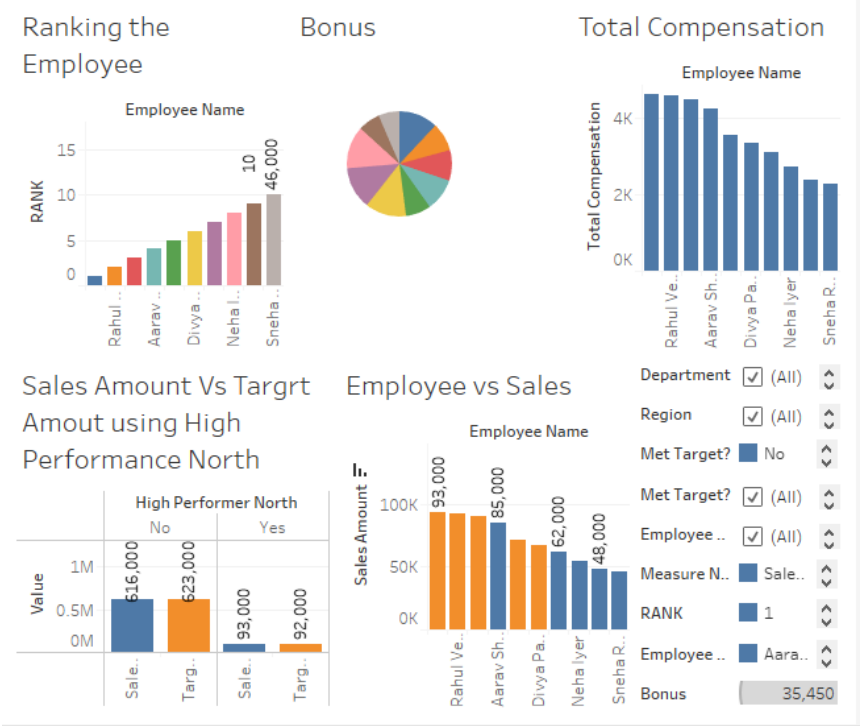
Step 24: Drag Bonus to Rows.

Step 25: Change the chart to a Pie Chart.

Step 26: Create a Dashboard.

Step 27: Save the process and Stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:04	Joining Table

AIM:

To perform various data visualizations and Joining table calculations in Tableau.

PROCEDURE:

Step 1: Start the process.

Step 2: Open Tableau Public. In the Connect pane, click on Text File, then select the flight_ex4 .csvdataset. Inspect the dataset to check its values and data types.

Step 3: Go to Data Source window ->Click on add button in the left pane,click on Text File, then select the second dataset airline_ex4.csv dataset. Inspect the dataset to check its values and data types.

Step 4: Drag the second CSV file (airline_ex4.csv) from the "Files" pane and drop it onto the canvas next to the first CSV file (flight_ex4.csv) to establish a relationship.

Step 5: Double-click the first CSV file (flight_ex4.csv) to open the physical layer, then drag the second CSV file (airline_ex4.csv) onto the canvas to create a join, which is visually represented by a Venn diagram icon.

Step 6: Click the join icon (Venn diagram) to open the configuration pane, then select the required join type and verify the join clauses to finalize the data merge.

Step 7: Click sheet 1 ->Drag the Airline dimension to the Rows shelf and drag the Flight ID measure to the Columns shelf, then ensure the aggregation of Flight ID is set to Count.

Step 8: Click sheet 2 ->A horizontal bar chart was created by dragging the Airline dimension to the Rows shelf and the Source field to the Columns shelf, automatically creating a measure of CNT(Source) (Count of Source) to show the flight count by origin airport for each airline.

Step 9: Click sheet 3 ->Open a new calculated field named Delayed was created using the formula IF [Status] = "Delayed" THEN "Delayed" END to identify delayed flights, and this field was then used to create a bar chart visualizing the count of airlines with delayed flights.

Step 10: Click sheet 4 ->A vertical bar chart was created by placing the Airline dimension on the Columns shelf, the CNT(Delayed) measure on the Rows shelf, and then placing a filter on the worksheet to show only records where the Delayed calculated field equals 'Yes' to visualize the count of delayed flights per airline.

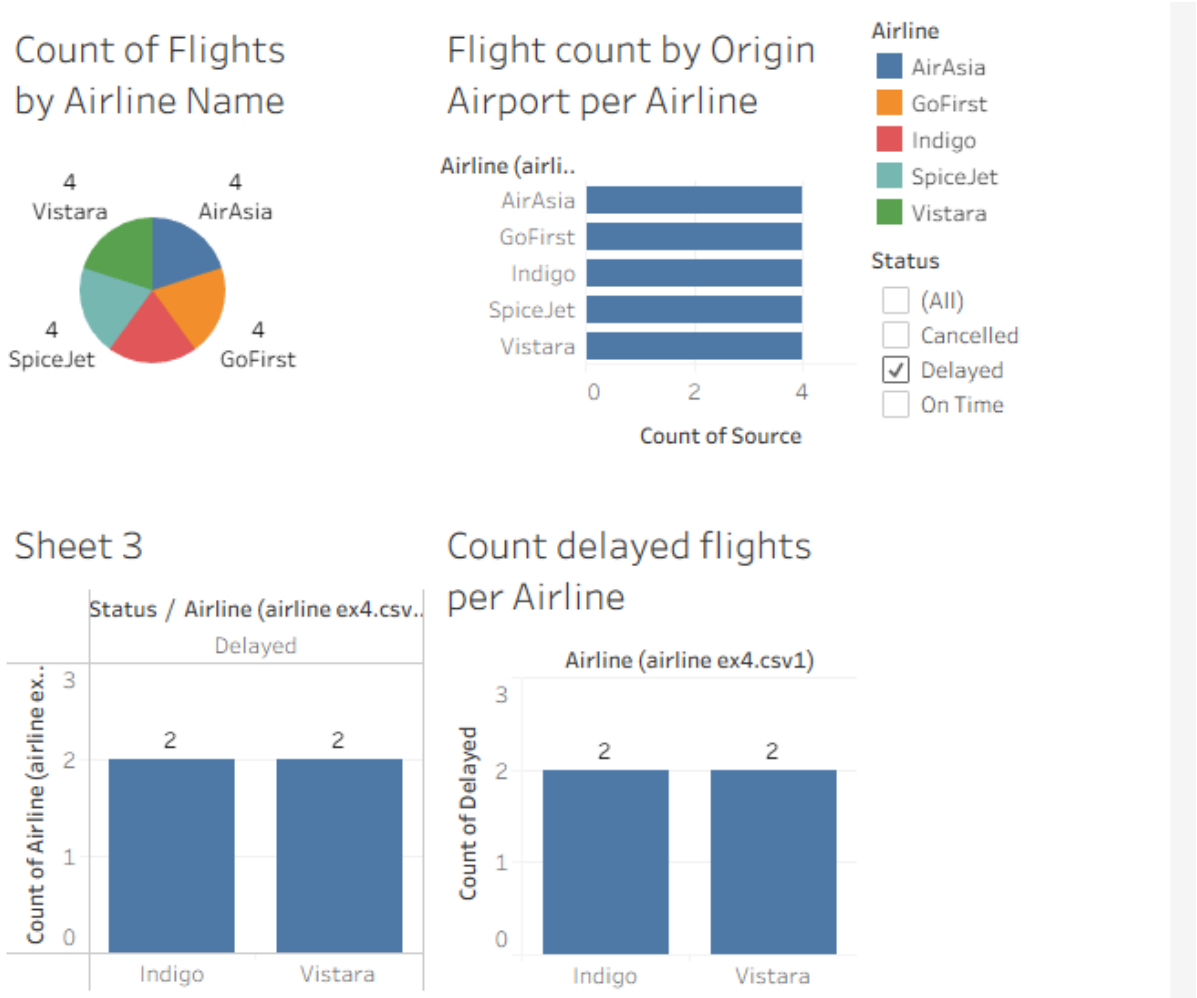
Step 11: Go to Dashboard 1 and place all four sheets accordingly.

Step 12: Save the project in the following location:

C:\Users\ADMIN\Documents\My Tableau Repository\Workbooks.

Step 13: View the output and stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:05	Types of Sorting

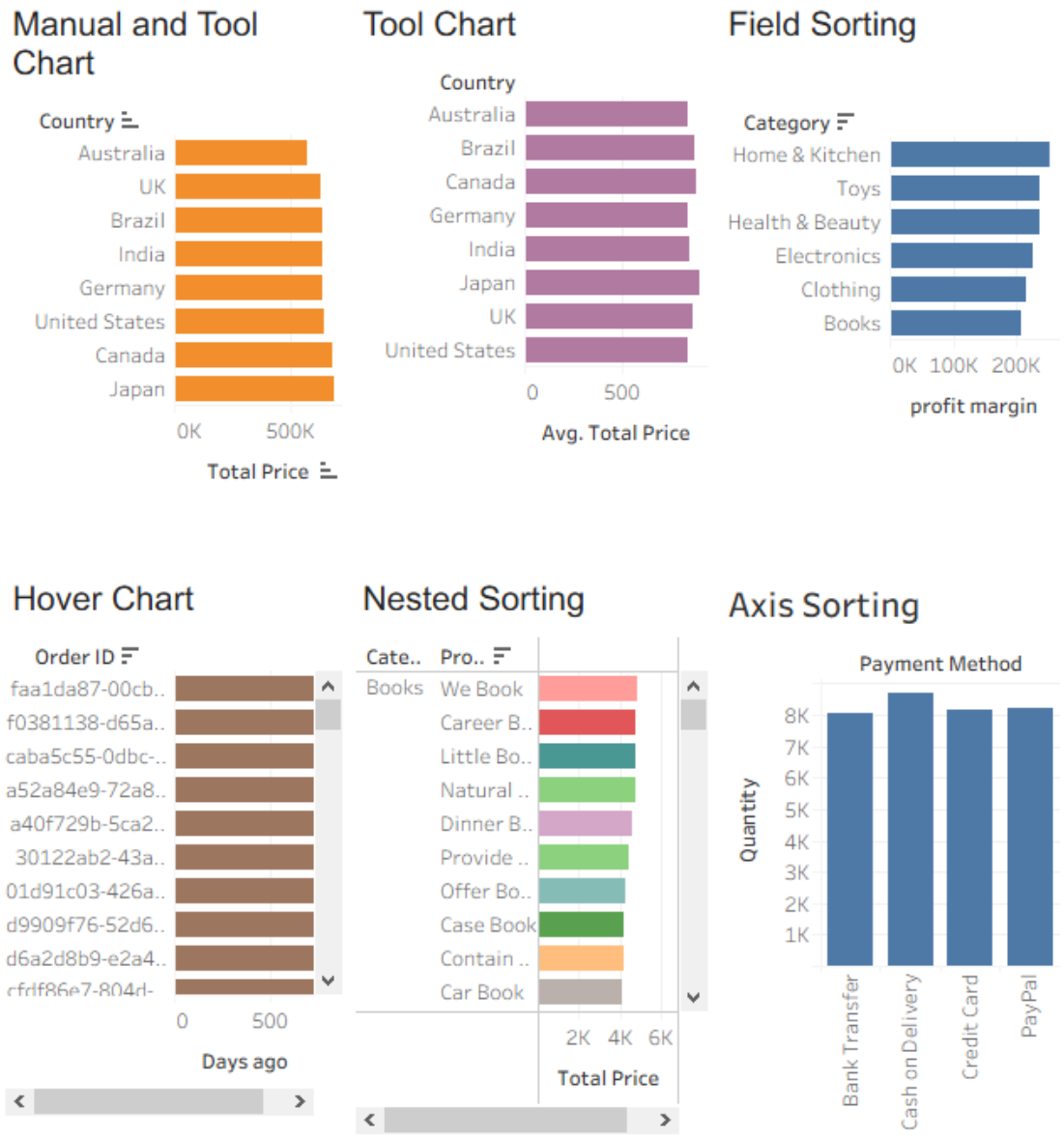
AIM:

To visualize different types of sorting in data analysis using Tableau.

PROCEDURE:

Step 1: Start the process.
Step 2: Open **Tableau Public** from your system.
Step 3: In the *Connect* pane, click on **Text File** and select the dataset (e.g., common_project_metrics.csv).
Step 4: Check the dataset preview to verify all fields and data types.
Step 5: Go to a new worksheet and rename it as **Manual Sort**.
Step 6: For **Manual Sort – Type 1 (Drag and Drop):**
Drag the required category or label and drop it in the desired position to manually reorder the data.
Step 7: For **Manual Sort – Type 2 (By Selecting Field):**
Click the field drop-down → choose **Sort** → select **Manual** → rearrange values manually → click **OK**.
Step 8: Go to a new worksheet and rename it as **Toolbar Sort**.
Step 9: Click the **Sort Ascending (A→Z)** or **Sort Descending (Z→A)** icon from the toolbar to arrange values automatically.
Step 10: Create another sheet and rename it as **Sort by Field**.
Step 11: Click the field drop-down → select **Sort by Field** → choose another field (like *Sales* or *Profit*) → select **Ascending/Descending** order.
Step 12: Create a new sheet named **Hover Sort**.
Hover your mouse over the field header or axis → click the **sort icon (↑ or ↓)** to quickly change order.
Step 13: Go to a new sheet and rename it as **Axis Sort**.
Click directly on the axis label → select **Sort** → choose criteria (by name, by value, or manual) → confirm sorting order.
Step 14: Add labels or color if needed to visualize the sorted order clearly.
Step 15: Save the Tableau workbook

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:06	**Tableau Data Charts** (Bar, Line, Pie, Scatter Plot — Same Dataset)

AIM:

To perform data visualization in Tableau to analyze the total revenue generated by each state.

PROCEDURE:

Step 1: Open Tableau Public and connect to the dataset using the Text File option.
Bar Chart
Step 2: Go to Sheet 1.
Step 3: Drag **State** to the **Columns** and **Revenue** to the **Rows**.
Step 4: Change the aggregation of Revenue to **SUM(Revenue).**
Step 5: From the **Show Me** panel, select the **Bar Chart** visualization.
Step 6: Add labels or tooltips to display **State** and **Revenue** values.
Step 7: Sort the bars in descending order and rename the chart as **"Revenue by State"**.

Line Chart
Step 8: Go to Sheet 2.
Step 9: Drag **Month** to the **Columns** and **Revenue** to the **Rows**.
Step 10: Change the Month field to **MONTH** and select **Line Chart** from the **Show Me** panel.
Step 11: Drag **Category** to the **Color** mark to differentiate categories.
Step 12: Add labels or tooltips to show **Category** and **Revenue** values.
Step 13: Rename the chart title as **"Monthly Revenue by Category."**

Pie Chart

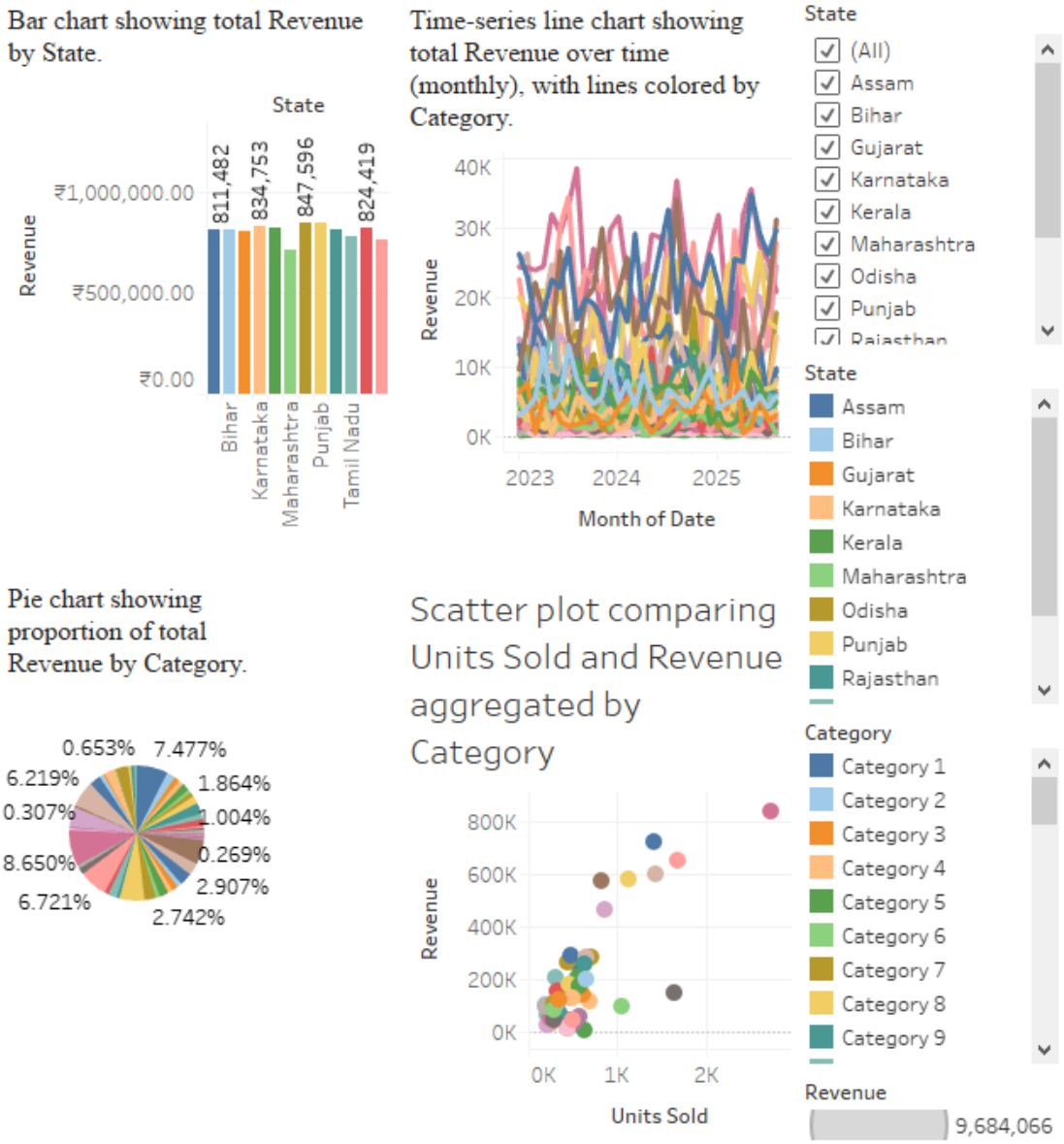
Step 14: Go to Sheet 3.
Step 15: Drag **Category** to the **Color** mark and **Revenue** to the **Angle** mark.
Step 16: Select the **Pie Chart** visualization from the **Show Me** panel.
Step 16: Right-click **SUM(Revenue)** and apply **Quick Table Calculation → Percentage of Total.**
Step 17: Add labels to show **Category names** and **Percentages**.
Step 18: Rename the chart title as **"Proportion of Total Revenue by Category."**

Scatter Plot

Step 19: Go to Sheet 1.
Step 20: Drag **Units Sold** to the **Columns** and **Revenue** to the **Rows**.
Step 21: Select the **Scatter Plot** visualization from the **Show Me** panel.
Step 22: Drag **Category** to the **Color** and **Label** marks to display category names.
Step 23: Adjust the size of the data points for better visibility.

Step 24: Rename the chart title as **"Units Sold vs Revenue (by Category)"**.
Step 25: View the output and stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:07	Pareto and Dual Axis

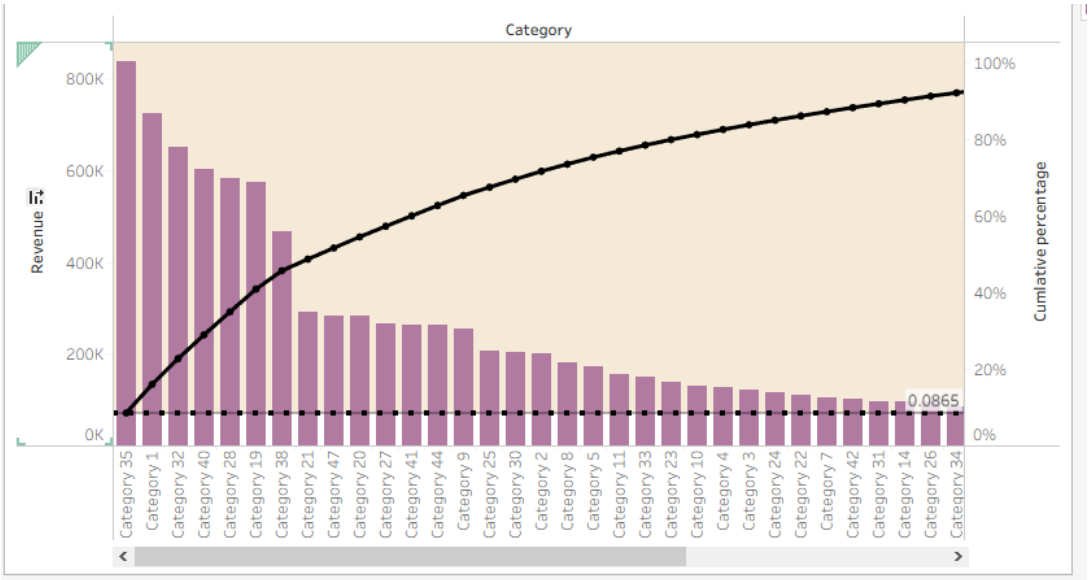
AIM:

To visualize a Pareto Chart in Tableau showing the cumulative percentage of Revenue across Categories using a dual-axis visualization.

PROCEDURE:

Step 1: Open **Tableau Public** and connect to the dataset using the **Text File** option.
Step 2: Go to **Sheet 1**.
Step 3: Drag **Category** to the **Columns** shelf.
Step 4: Drag **Revenue** to the **Rows** shelf.
Step 5: Click the dropdown on **Revenue** and select **Sort → Descending** to arrange categories from highest to lowest revenue.
Step 6: Create a new **Calculated Field** named *Cumulative Percentage* with the formula:
RUNNING_SUM(SUM([Revenue])) / WINDOW_SUM(SUM([Revenue]))
Step 7: Drag the new field **Cumulative Percentage** to the **Rows** shelf, next to Revenue.
Step 8: Right-click on the **Cumulative Percentage** axis and select **Dual Axis** to combine both charts.
Step 9: From the **Marks** card, change **Revenue** to **Bar** and **Cumulative Percentage** to **Line** for a dual representation.
Step 10: Right-click the right-side Y-axis (**Cumulative Percentage**) and select **Add Reference Line** to highlight key percentage thresholds.
Step 11: Adjust chart colors, bar widths, and line thickness for clarity and better readability.
Step 12: Rename the chart title as **"Pareto Chart (Dual Axis)"**, view the output, and stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:08	Motion Chart

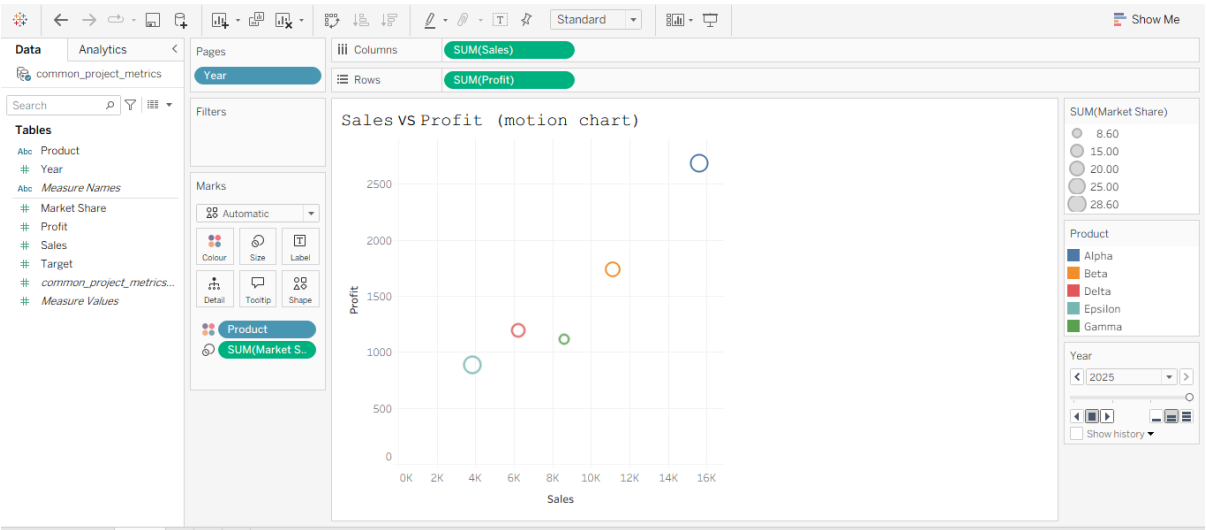
AIM:

To visualize a motion chart in Tableau.

PROCEDURE:

Step 1: Open Tableau Public and connect to the dataset using the **Text File** option.
Step 2: Go to **Sheet 1**.
Step 3: Drag **Sales** to the **Columns** shelf.
Step 4: Drag **Profit** to the **Rows** shelf.
Step 5: Drag **Product** to the **Color** mark to differentiate products by color.
Step 6: Drag **Market Share** to the **Size** mark to represent bubble size by market share.
Step 7: Drag **Year** to the **Pages** shelf to enable animation over time.
Step 8: From the **Marks** card, select **Circle** as the mark type.
Step 9: Use the **Play** button in the Pages shelf to animate the bubbles, showing changes in Sales, Profit, and Market Share across years.
Step 10: Adjust the size and color intensity of bubbles for better visibility.
Step 11: Rename the chart title as **"Sales vs Profit (Motion Chart)"**.
Step 12: View the animated output and stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:09	Lollipop or Dot Chart

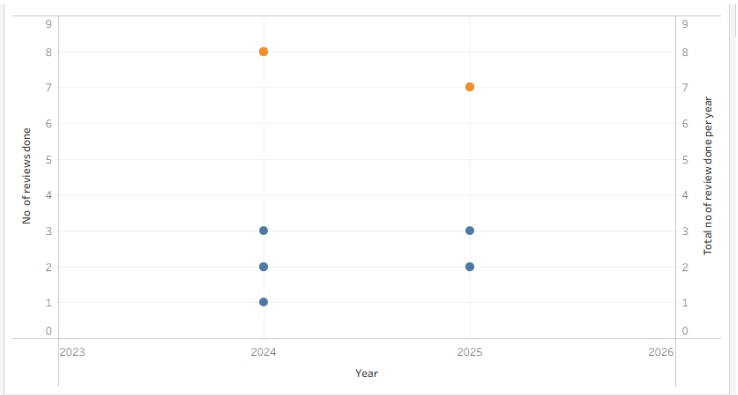
AIM:

To visualize lollipop or Dot Chart in Tableau showing the total number of reviews done per year using a dual-axis view.

PROCEDURE:

Step 1: Open **Tableau Public** and connect to the dataset using the **Text File** option.
Step 2: Go to **Sheet 1**.
Step 3: Drag **Year** to the **Columns** shelf.
Step 4: Drag **No of reviews done** to the **Rows** shelf.
Step 5: Create a **Calculated Field** named *Total no of review done per year* with the formula:
{ FIXED [Year] : SUM([No of reviews done]) }
Step 6: Drag this new calculated field **Total no of review done per year** to the **Rows** shelf next to the existing measure.
Step 7: Right-click on the **Total no of review done per year** axis and select **Dual Axis** to display both measures together.
Step 8: From the **Marks** card, change both measures to **Circle** type to form a dotted chart.
Step 9: Assign different colors to **No of reviews done** and **Total no of review done per year** for better distinction.
Step 10: Synchronize the axes and adjust the size of dots for better visualization.
Step 11: Add **Name of the journal reviewer** to the **Detail** or **Tooltip** to display additional information on hover.
Step 12: Rename the chart title as **"Dotted Chart – Total Reviews Done per Year"**, view the output, and stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:10	Gantt Bar Chart

AIM:

To create a Gantt chart in Tableau to visualize project tasks along with their start dates, durations, and progress status.

PROCEDURE:

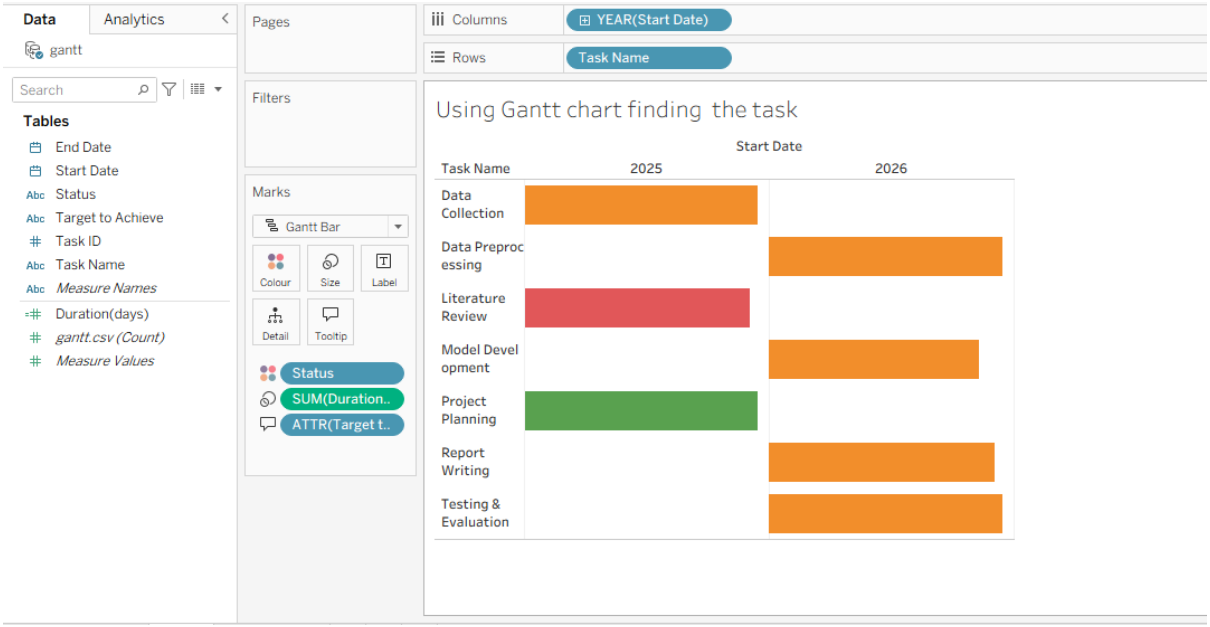
Step 1: Start the process.

Step 2: Open Tableau Public. In the *Connect* pane, click on Text File and import the dataset. Inspect the dataset to ensure all fields such as Task Name, Start Date, End Date, Duration, and Status are correctly recognized.
Step 3: Go to Sheet 1.

Step 4: Drag Start Date to the Columns.
Step 5: Drag Task Name to the Rows.
Step 6: From the Marks card, change the mark type to Gantt Bar.
Step 7: Drag Duration (in days) to the Size mark to represent the length of each task on the Gantt chart.
Step 8: Drag Status to the Color mark to visually differentiate tasks based on their progress or status.
Step 9: Optionally, add Target or End Date to the Tooltip to display additional information when hovering over each task.
Step 10: Adjust the bar sizes and colors for better readability.
Step 11: Rename the chart title as "Using Gantt Chart Finding the Task".
Step 12: Save the workbook at the following location:
C:\Users\ADMIN\Documents\My Tableau Repository\Workbooks

Step 13: Stop the Process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:11	Cross Tab Chart

AIM:

To visualize a Cross Table Chart (Text Table) in Tableau showing the Sales and Profit of different Product Categories across various Regions, and to analyze the regional performance for each category.

PROCEDURE:

Step 1: Start the process.

Step 2: Open Tableau Public. In the Connect pane, click on Text File, then select the product_per dataset. Inspect the dataset to check its values and data types.

Step 3: Go to Sheet 1 -> Drag Product category to the Rows pane and Region to the Columns pane.

Step 4: Drag Sales and Profit to the Text field in the Marks pane.

Step 5: Drag Sales to color in the Marks pane for visualization.

Step 6: Change Automatic to Square in Marks Pane.

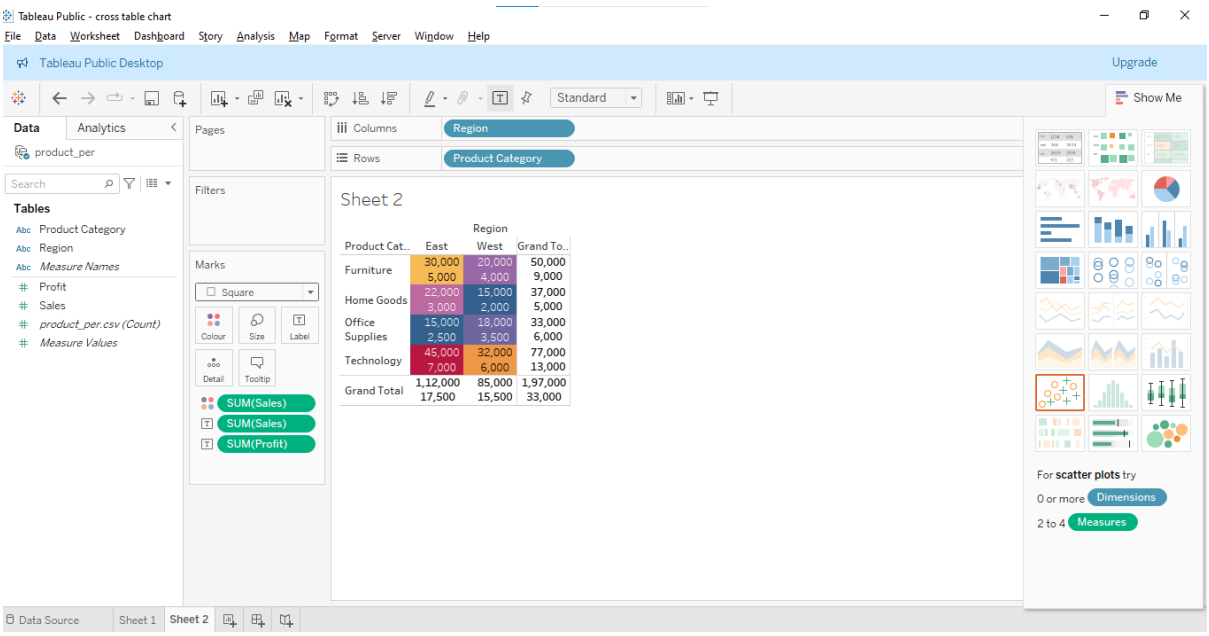
Step 7: Enable Row and Column Grand Totals from the Analysis → Totals menu and select both show row grand total and show column grand total.

Step 8: Rename the sheet.

Step 9: Save the workbook.

Step 10: Stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:12	Bump Chart

AIM:

To create a Bump chart in Tableau showing the Sales Rank of Products over different Years to analyze how product performance changes over time.

PROCEDURE:

Step 1: Start the process.

Step 2: Open Tableau Public. In the Connect pane, click on Text File, then select the product_per dataset. Inspect the dataset to check its values and data types.

Step 3: Go to Sheet 1 -> Drag Sales Rank twice to the Rows pane and Year to the Columns pane.

Step 4: Right click the Sales Rank 2 and Select Dual Axis.

Step 5: Go to Sales Rank 1 in the Marks pane → change Automatic to Line.

Step 6: Drag Product to Color and Label and Drag Sales Rank to Label.

Step 7: Go to Sales Rank 2 in the Marks pane → change Automatic to circle.

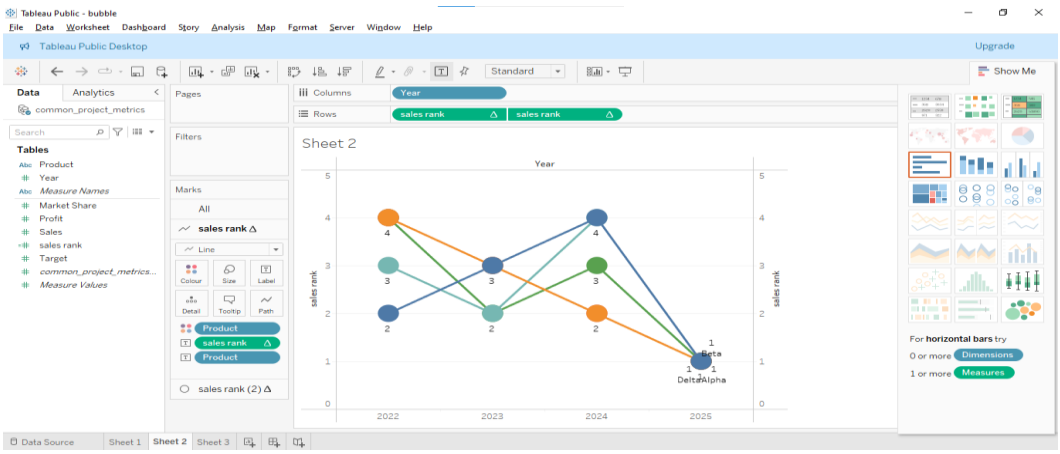
Step 8: Drag Product to Color.

Step 9: Rename the sheet.

Step 10: Save the Workbook.

Step 11: Stop the Process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:13	Bubble Chart, Waterfall, Area Chart

AIM:

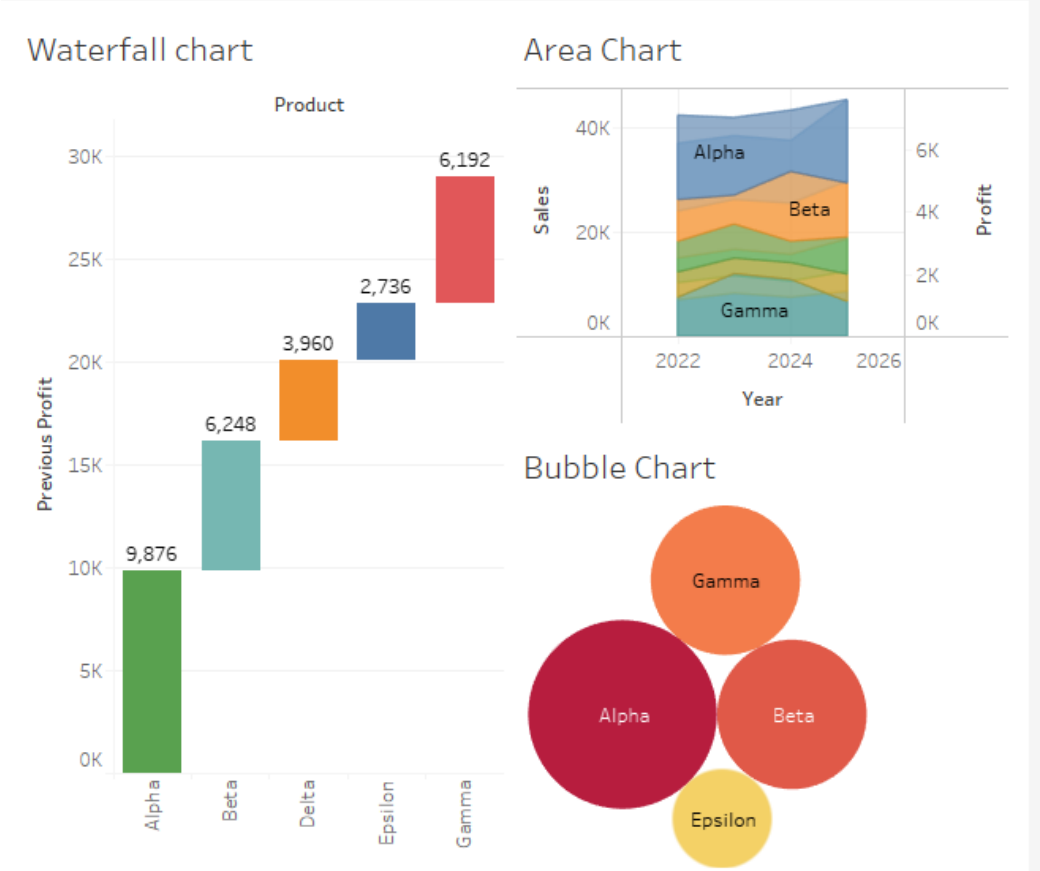
To data visualization in Tableau by creating **Waterfall Chart, Area Chart, and Bubble Chart** using the **common_project_metrics** dataset.

PROCEDURE:

Step 1: Start the process.

Step 2: Open **Tableau Public**. In the **Connect** pane, click on **Text File**, then select the **common_project_metrics** dataset. Inspect the dataset to check its values and data types.**Step 3:** Go to a new sheet and rename it as **Waterfall Chart**.
Step 4: Drag **Product** to the *Columns* and **SUM(Profit)** to the *Rows* to create a bar chart.
Step 5: Change the mark type to **Gantt Bar** using the *Marks* card.
Step 6: Create a calculated field named **Previous Profit** using the formula:
RUNNING_SUM(SUM([Profit])) - SUM([Profit])
Step 7: Drag **Previous Profit** to *Rows* (replace the existing SUM(Profit))and **SUM(Profit)** to *Size* and *Color* for variation.
Step 8: Add labels and sort **Products** in descending order based on profit.
Step 9: Go to a new sheet and rename it as **Area Chart**.
Step 10: Drag **Year** to the *Columns* pane and **Sales** to the *Rows*.
Step 11: Change the mark type to **Area** in the *Marks* card.
Step 12: Drag **Product** to *Color* to represent multiple products with different shades.
Step 13: Drag **Profit** to *Rows* again, right-click the second axis, and choose **Dual Axis.**
Step 14: Drag product to filter.
Step 15: Go to a new sheet and rename it as **Bubble Chart.**
Step 16: Drag **SUM(Sales)** to the *Columns* pane and **SUM(Profit)** to the *Rows* pane.
Step 17: From the *Show Me* panel, select **Circle (Bubble)** chart type.
Step 18: Drag **Sales** to *Color* and **Product** to *Label* to identify each product.
Step 19: Drag **SUM(Profit)** to *Size* to represent bubble size based on total sales.
Step 20: Save the project in the following location:
C:\Users\ADMIN\Documents\My Tableau Repository\Workbooks
Step 21: View the output and stop the process.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:14	Heat Map

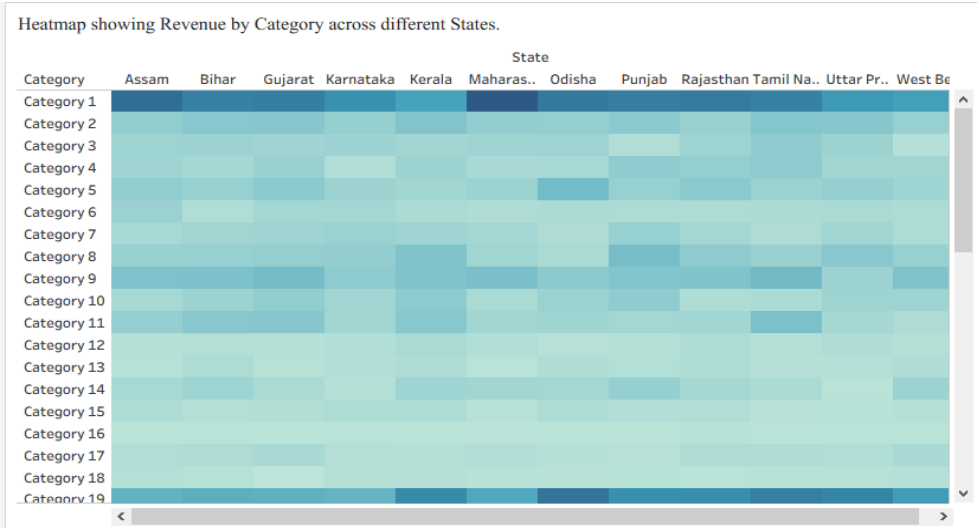
AIM:

To create a Heat Map in Tableau Public to visualize revenue distribution across different States and Categories, and to apply animation for better data interpretation.

PROCEDURE:

Step 1: Start the process.
Step 2: Open Tableau Public. In the Connect pane, click on Text File, then select the pareto_sales_10000.csv dataset. Inspect the dataset to check its values and data types.
Step 3: Go to Sheet 1 → Drag State to the Column pane and Category to the Row pane.
Step 4: Drag Revenue to the Color option in the Marks card.
Step 5: In the Marks card, change the mark type from Automatic to Square.
Step 6: Click the Color tab in the Marks card and edit the color palette as desired.
Step 7: Double-click the title and rename it as "Heat Map Chart."
Step 8: To add animation, go to the top menu and click Format → Animations. In the Animations pane, turn Workbook Animation: ON, choose Animation Style: Simultaneous, and set the Duration (for example, 1 second). Then, apply a filter on State to see the animation effect.

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.

EX NO:15	Tree Map

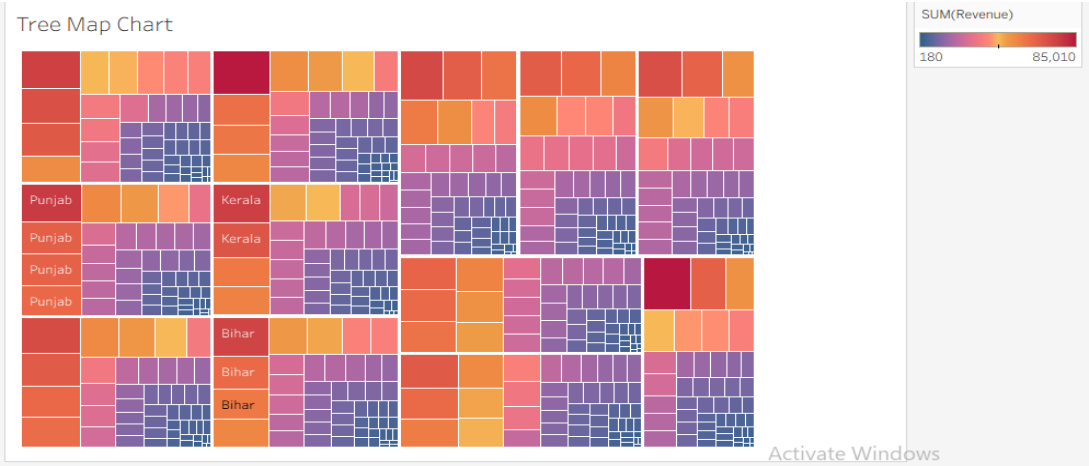
AIM:

To visualize a Tree Map in Tableau Public to represent revenue by Category and State, using size and color to compare values effectively.

PROCEDURE:

Step 1: Start the process.
Step 2: Open Tableau Public. In the Connect pane, click on Text File, then select the dataset. Inspect the dataset to check its values and data types.
Step 3: Go to Sheet 1 -> Drag Category to the Rows pane
Step 4: Drag State to the Detail option in the Marks card.
Step 5: Drag Revenue to the Size and Color option in the Marks card and edit your color.
Step 6: In the Show me button, click the treemap.
Step 7: Double-click the title and rename it as "Tree Map Chart."

OUTPUT:



RESULT:

Thus, the above program is executed successfully and verified.